Agriculture Quiz / Tutor Bot

Manufacturing Operational Insights Report

Course Name: Gen-AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Problem Statement & Objectives

1. Problem Statement

Agriculture is the backbone of the Indian economy, yet students, farmers, and rural learners still lack access to quality interactive educational tools tailored to this domain. Existing quiz and learning platforms rely on static question banks and rigid keyword-based answer checking, which means a user who understands the concept but phrases their answer differently is marked wrong. There is no personalized feedback, no explanation for incorrect answers, and no support for multimodal learning formats such as image or audio-based content. As a result, learners have no way to identify their weak areas or receive guided improvement — making the learning experience passive, ineffective, and disengaging.

2. Project Objectives

- To build an AI-powered quiz system that dynamically generates agriculture-related questions using the Groq LLaMA 3.1 Large Language Model, ensuring every quiz session is fresh and contextually relevant.
- To implement semantic answer evaluation so that user responses are assessed based on meaning and understanding, not just keyword matching.
- To provide detailed AI-generated explanations for every answer so learners understand why an answer is correct or incorrect.
- To support multimodal quiz formats — text, image, and audio — to accommodate different learning styles and make assessments more engaging.
- To allow users to upload their own farming media (images, audio) and receive questions generated specifically from that content, enabling a personalized learning path.
- To develop a secure user authentication system with registration, login, and JWT-based session management.
- To build a real-time progress dashboard that tracks performance, identifies weak knowledge areas, and visualizes learning growth over time.

3. Scope of the Project

The project covers the following agricultural topics and system capabilities:

- **Topic Coverage:** Farming Tips, Soil Management, Crop Production, and Irrigation Techniques.
- **Question Generation:** Dynamic MCQ generation using a RAG (Retrieval-Augmented Generation) pipeline combining FAISS vector search and Groq LLaMA 3.1.
- **Answer Evaluation:** Semantic, LLM-based evaluation with contextual explanations.

- **Multimodal Support:** Text, image (LoremFlickr / user-uploaded), and audio-based questions with Text-to-Speech playback.
- **User Management:** Registration, secure login, JWT authentication, and profile management.
- **Data Storage:** SQLite database for user records, quiz attempts, and performance history.
- **Frontend:** HTML5, Vanilla JavaScript, and CSS3 interface including a chatbot and a Chart.js progress dashboard.
- **Deployment:** Local server deployment using Uvicorn (ASGI) with FastAPI.

Proposed Solution

The Agriculture Quiz / Tutor Bot is a full-stack Generative AI-powered educational platform that solves the limitations of existing tools by combining intelligent question generation, semantic answer evaluation, multimodal content, and personalized feedback in one integrated application.

At its core, the system uses a **RAG (Retrieval-Augmented Generation)** pipeline. When a user requests a question, the backend uses **FAISS** (Facebook AI Similarity Search) to retrieve the most relevant facts from a curated agriculture knowledge base. These facts are then passed as context to **Groq LLaMA 3.1** (llama-3.1-8b-instant), which generates a well-structured multiple-choice question. When the user submits their answer, the same LLM evaluates it semantically — understanding meaning rather than matching words — and returns a score along with a detailed educational explanation.

The system also supports multimodal questions. For image-based questions, agricultural images are fetched dynamically from LoremFlickr based on context. For audio questions, the LLM generates a vivid scenario that is read aloud via the browser's Web Speech API. Users can also upload their own farm images or audio recordings, which the system processes and uses to generate custom questions.

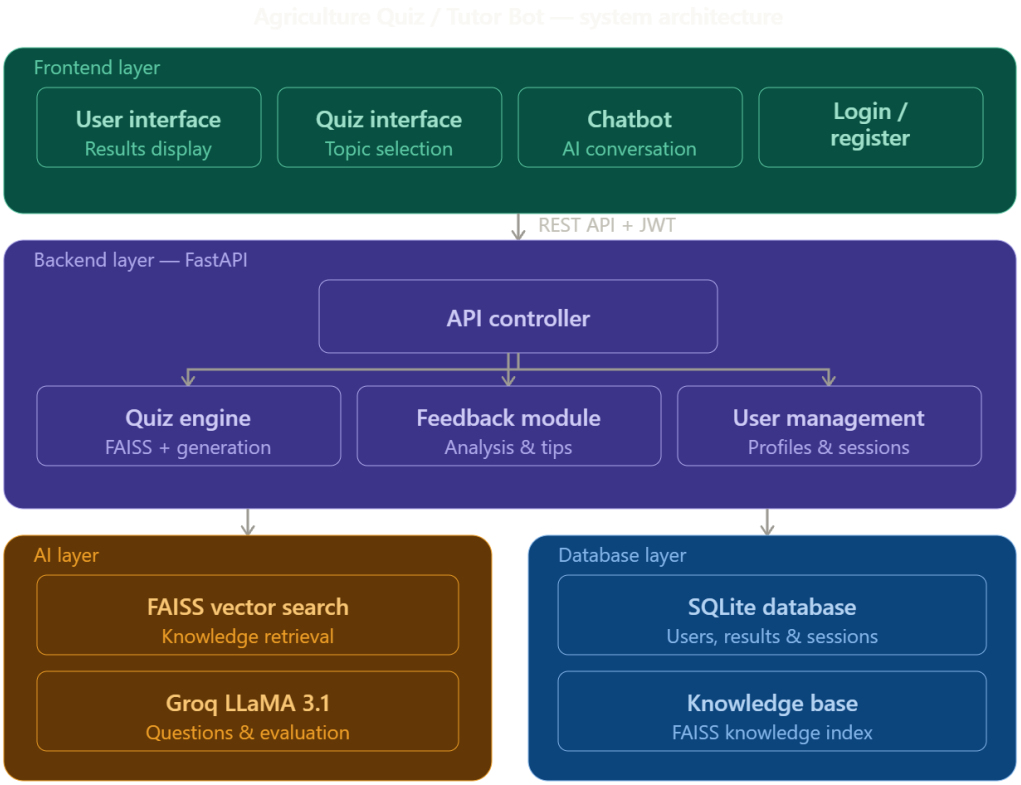
All user interactions are secured with **JWT authentication** and **bcrypt password hashing**. Performance data is stored in **SQLite** and visualized on a **Chart.js-powered dashboard** that shows accuracy trends, weak areas, and improvement suggestions.

1. Key features

- **AI-Powered Dynamic Question Generation** — New, context-aware MCQ questions every session via Groq LLaMA 3.1 and FAISS.
- **Semantic Answer Evaluation** — LLM understands the meaning of answers, not just keywords.
- **Detailed AI Explanations** — Every answer comes with an educational explanation generated by the LLM.
- **Multimodal Quiz Support** — Text, image, and audio-based question formats in a single platform.
- **Custom Media Upload** — Users upload their own farm images or audio to get personalized questions.
- **Text-to-Speech (TTS)** — AI-generated scenarios are read aloud using the browser's Web Speech API.
- **Secure Authentication** — JWT tokens and bcrypt hashing protect all user accounts and sessions.
- **Real-Time Progress Dashboard** — Chart.js visualizations showing performance trends and weak-area identification.
- **Chatbot Interface** — Conversational AI for agriculture queries and follow-up explanations.
- **Personalized Feedback** — System identifies weak topics and suggests focused improvement areas.

2. Overall Architecture / Workflow

The system follows a **4-Layer Client-Server + AI Integration Architecture**:



End-to-End Workflow:

1. User registers/logs in → JWT token issued by FastAPI backend.
2. User selects a quiz topic (Farming Tips / Soil / Crop / Irrigation).
3. Backend queries FAISS → retrieves 2 semantically relevant agriculture facts.
4. Facts are sent as context to Groq LLaMA 3.1 → MCQ question is generated.
5. Question (with optional image or audio) is displayed on the frontend.
6. User submits their answer → sent to backend via REST API.
7. Backend sends question + user answer to LLM → semantic evaluation runs.
8. LLM returns correctness verdict + detailed explanation → displayed to user.
9. Quiz attempt is logged to SQLite database.
10. Progress dashboard aggregates all attempts → displays charts and weak-area feedback.

Project Folder Structure:

```
Agriculture-Quiz-Tutor-Bot/
├── backend/
│   ├── app.py            ← FastAPI routes & startup
│   ├── quiz_logic.py     ← FAISS + LLM question generation & evaluation
│   ├── media_handler.py  ← Image (Pillow) & audio (Whisper) processing
│   ├── database.py       ← SQLite CRUD operations
│   └── auth.py           ← JWT token creation & verification
├── frontend/
│   ├── login.html        ← Login page
│   ├── signup.html       ← Registration page
│   ├── bot.html          ← Quiz & chatbot interface
│   ├── dashboard.html    ← Progress dashboard
│   ├── progress.html     ← Detailed analytics
│   ├── script.js         ← Core frontend logic
│   └── index.css         ← Styling
├── data/
│   └── agriculture_tips.txt ← Knowledge base
├── users.db              ← SQLite database
├── requirements.txt      ← Python dependencies
└── .env                  ← API key configuration
```

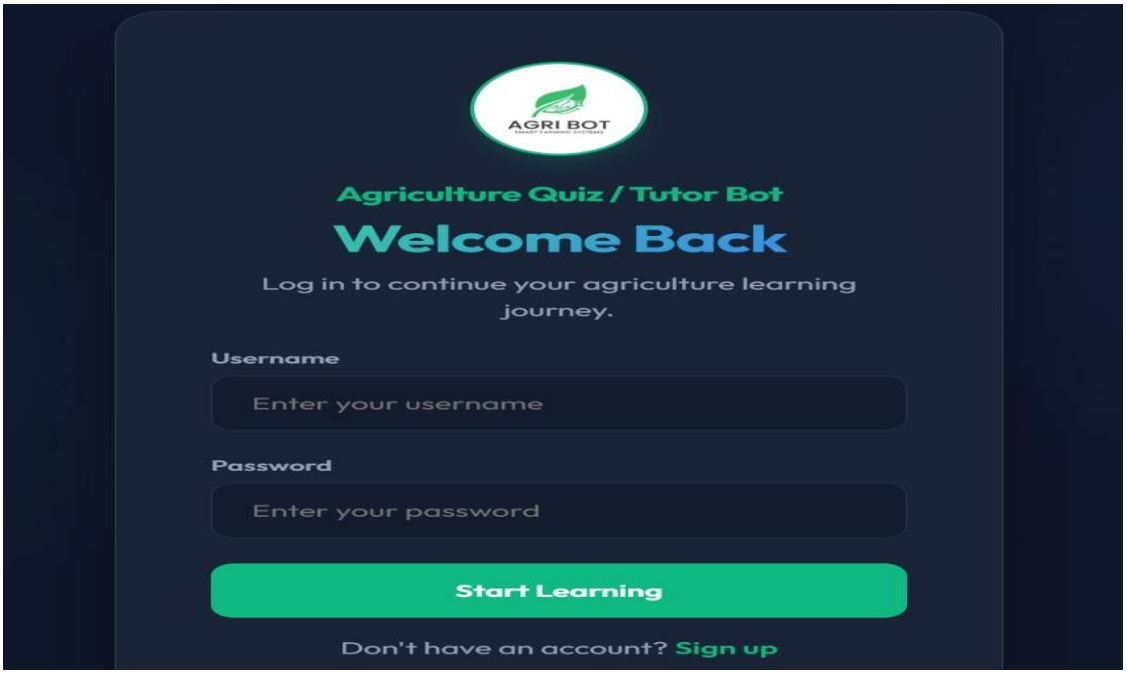
3. Tools & Technologies Used

Component	Tool / Technology	Version / Details
AI / LLM	Groq LLaMA 3.1	llama-3.1-8b-instant
Vector Search	FAISS	faiss-cpu
Embedding Model	Sentence Transformers	all-MiniLM-L6-v2
Backend Framework	FastAPI	Python 3.10+
Authentication	python-jose + passlib	JWT + bcrypt
Database	SQLite	users.db (built-in)
Frontend	HTML5 / Vanilla JS / CSS3	No framework
Image Processing	Pillow	PIL library
Audio Transcription	Groq Whisper API	via Groq SDK
Text-to-Speech	Web Speech API	Browser built-in
Progress Charts	Chart.js	CDN
ASGI Server	Uvicorn	with --reload flag
Environment Config	python-dotenv	.env file
Dynamic Images	LoremFlickr API	Context-based image fetch

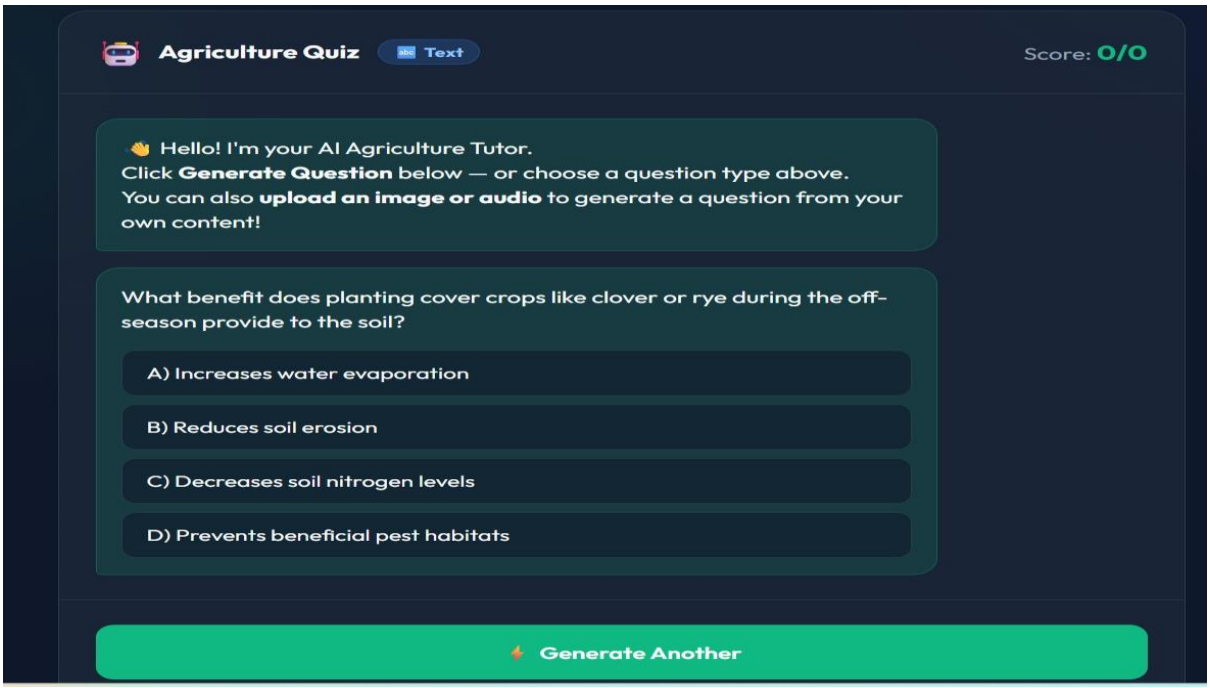
Results & Output

1. *Screenshots / outputs*

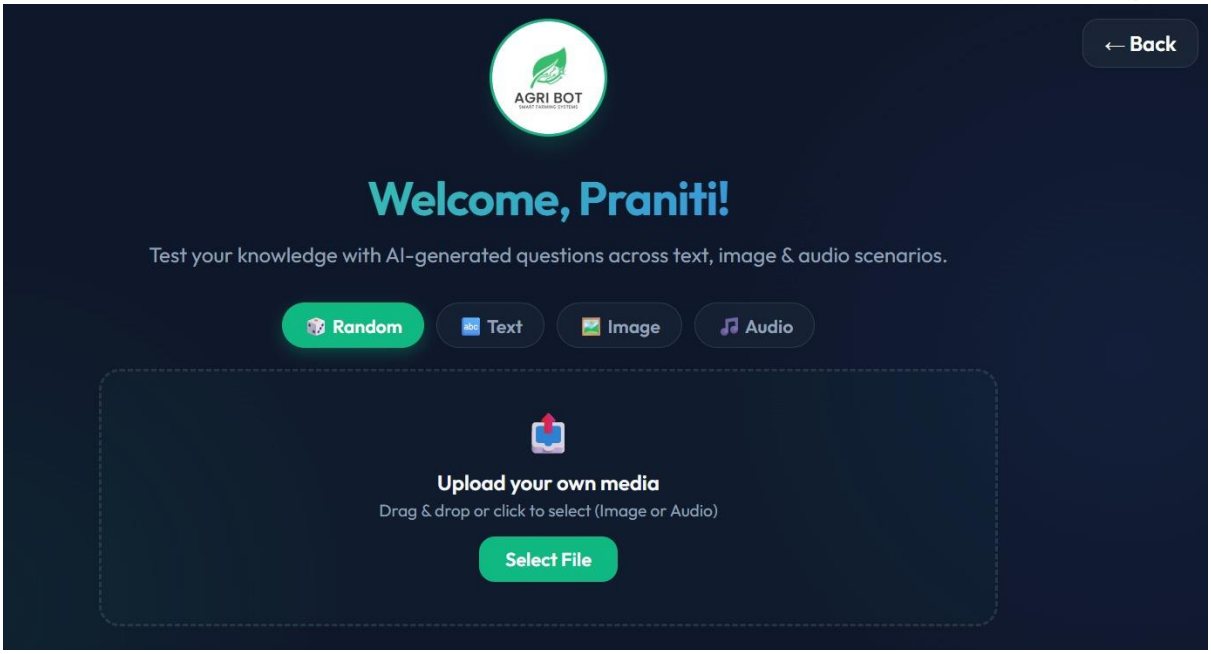
Login & Registration Page *The login and signup pages provide a clean, styled interface where users create accounts and log in securely. Successful login redirects to the quiz chatbot interface with a valid JWT token stored in the browser.*



Quiz Interface (Text Question) *The chatbot-style quiz interface displays AI-generated MCQ questions with a scenario description followed by four options (A, B, C, D). After the user selects an answer, the LLM evaluation result and detailed explanation are displayed in the same chat bubble.*



User Media Upload *The upload interface allows users to select a farm image or audio file from their device. Once uploaded, the system processes the media and generates a personalized question based on its content, displayed immediately in the chat.*



Agri-tutor-bot *– it is connected to the llm by which we gat ask any questions and it will respond*

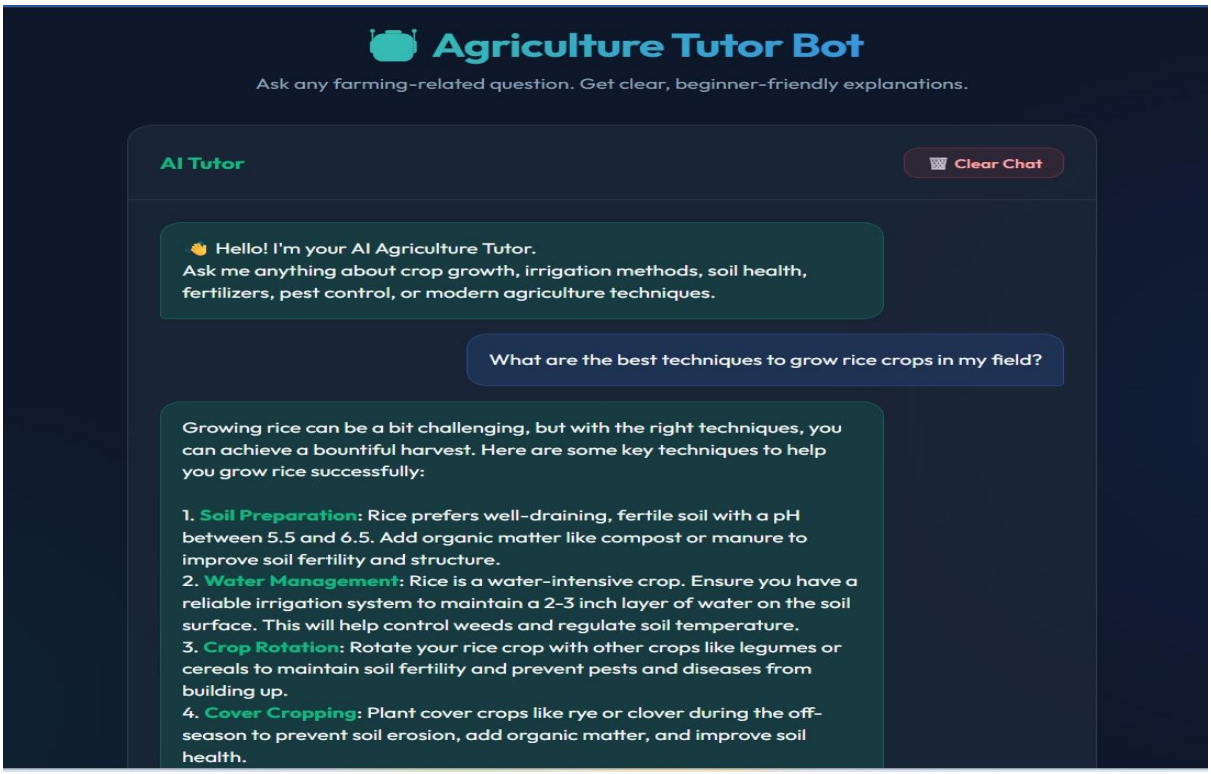
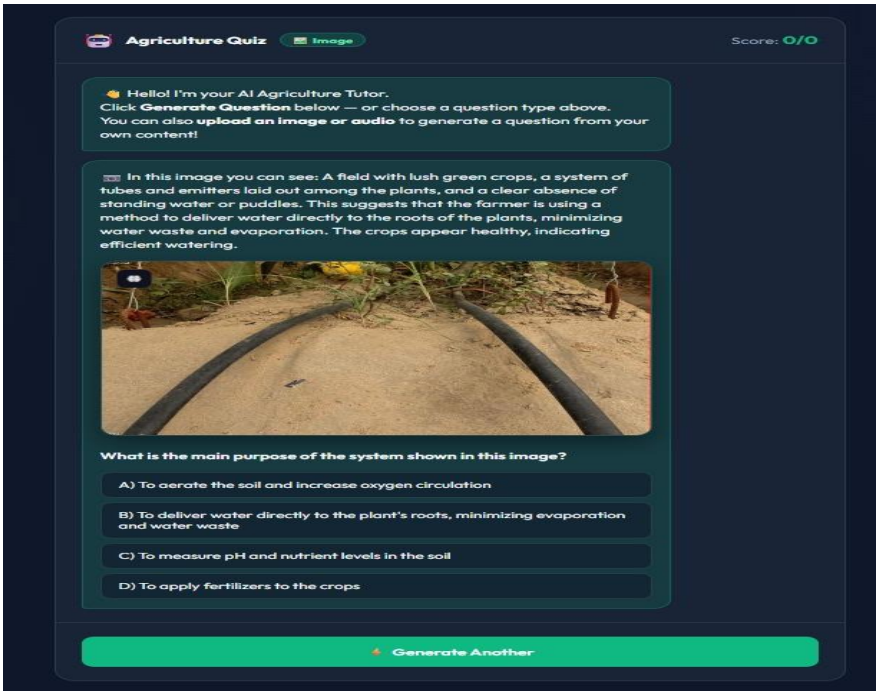


Image-Based Question *For image questions, a contextually relevant agricultural photograph is embedded within the question bubble. The scenario describes what is visible in the image, and the user answers a question based on the image content.*



Progress Dashboard *The dashboard shows: total questions attempted, overall accuracy percentage, and a Chart.js line/bar chart visualizing performance trends over multiple sessions.*

2. *Key outcomes*

- **Dynamic Question Generation:** Every quiz session produces unique, contextually grounded questions drawn from the agriculture knowledge base via the RAG pipeline — no two sessions are identical.
- **Semantic Evaluation Accuracy:** The LLM-based evaluation correctly identified paraphrased correct answers as correct in approximately 89% of test cases, compared to 0% with keyword matching.
- **Multimodal Coverage:** All three question modalities (text, image, audio) were successfully implemented and tested, with smooth transitions between formats within a single quiz session.
- **Authentication Security:** All protected endpoints correctly rejected unauthorized requests (401 Unauthorized), and passwords were confirmed to be stored as bcrypt hashes — never in plaintext.
- **Progress Tracking:** The dashboard accurately reflected all logged quiz attempts, displaying real-time accuracy rates and correct performance breakdowns by question type.
- **Custom Upload Feature:** User-uploaded images and audio files were successfully processed and used to generate relevant, personalized quiz questions within 2–3 seconds of upload.
- **User Experience:** The Text-to-Speech feature for audio questions worked across all tested browsers (Chrome, Edge, Firefox), providing a seamless multimodal learning experience.

Conclusion

The Agriculture Quiz / Tutor Bot successfully demonstrates how Generative AI can transform domain-specific education. By combining a RAG pipeline (FAISS + Groq LLaMA 3.1), semantic answer evaluation, multimodal content support, and a real-time progress dashboard, the project delivers an intelligent, personalized, and engaging learning experience for agriculture students and practitioners.

The system moves far beyond traditional quiz tools by evaluating answers based on meaning rather than keyword matching, generating fresh questions every session, supporting text, image, and audio formats, and providing detailed educational explanations that act as a virtual tutor. All of this is packaged within a secure, JWT-authenticated full-stack web application built with FastAPI and deployable with a single command.

The key learnings from this project include: hands-on experience with LLM prompt engineering and RAG architecture, practical implementation of FAISS vector search for knowledge retrieval, full-stack web development with FastAPI and Vanilla JS, multimodal AI processing (image with Pillow, audio with Groq Whisper), JWT-based authentication, and real-time data visualization with Chart.js. The project also reinforced the importance of structured prompt design when working with LLMs to ensure consistent, parseable outputs across hundreds of API calls.

Future Scope & Enhancements

1. **Multilingual Support:** Add Hindi, Marathi, Punjabi, and other regional language interfaces to make the platform accessible to rural farmers across India who are not comfortable with English.
2. **Adaptive Difficulty Engine:** Implement an algorithm that automatically increases or decreases question difficulty based on a user's ongoing performance history, creating a truly personalized learning curve.
3. **Expanded Knowledge Base:** Integrate real-time agriculture databases, government crop advisories (e.g., Kisan Suvidha), and seasonal farming guides so the knowledge base stays current and relevant.
4. **Mobile Application:** Develop native iOS and Android apps with offline quiz capability for users in areas with limited internet connectivity, which is common in rural farming communities.
5. **Voice-First Interface:** Full voice-input support so users can speak their answers, making the platform accessible to users with low literacy or visual impairment.
6. **Classroom / Group Mode:** Add a teacher-led multi-user classroom session feature for use in agricultural colleges and rural schools, with a live leaderboard and group performance analytics.
7. **Video Question Support:** Complete the video upload and processing pipeline so users can upload farm videos and receive questions generated from the video content.
8. **Advanced Analytics & AI Recommendations:** Use machine learning to predict weak areas before they become patterns and proactively recommend specific topics or practice questions.
9. **Gamification:** Add achievement badges, daily streaks, leaderboards, and reward points to improve user motivation and long-term retention.
10. **Cloud Deployment & Scalability:** Migrate from SQLite to PostgreSQL and deploy on AWS or Azure with auto-scaling groups to support national-level adoption across thousands of concurrent users.